GOVT COLLEGE WOMEN UNIVERSITY OF FAISALABAD

Department of Computer Science

Assignment no 1

OF

Analysis of Algorithm

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Section: BSCS (Morning C) 6th

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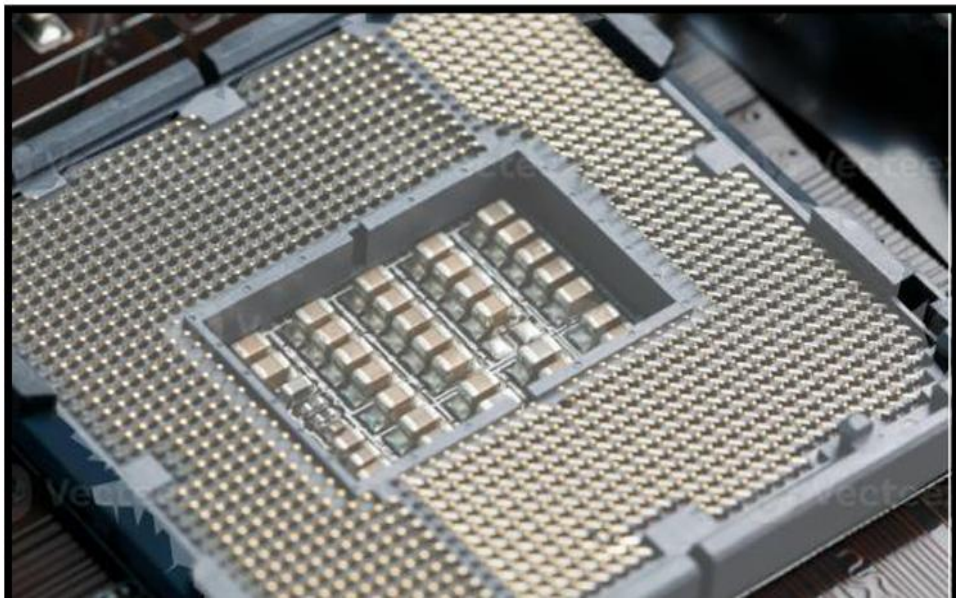
Motherboard:-

The motherboard is the main circuit board of a computer on which all important components such as the CPU, RAM, storage devices, and expansion cards are connected.
It allows communication between all hardware components and also supplies power to them.

Hardware Parts of Motherboard:-

1. **CPU Socket**
   Function: Used to install the processor
   Specifications:

   • Types: LGA, PGA, BGA
   • Intel and AMD use different sockets
   • Examples: LGA 1700, AM4, AM5



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2. **RAM Slots (DIMM Slots)**
   Function: Used to install RAM
   Specifications:

   • DDR Types: DDR3, DDR4, DDR5
   • Number of slots: 2 or 4
   • Capacity: 8GB, 16GB, 32GB per slot

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3. **Chipset**

Function: Controls communication between CPU, RAM, storage, and other devices
Specifications:

• Intel chipsets: H, B, Z series
• AMD chipsets: A, B, X series
• Features: Overclocking support, port support

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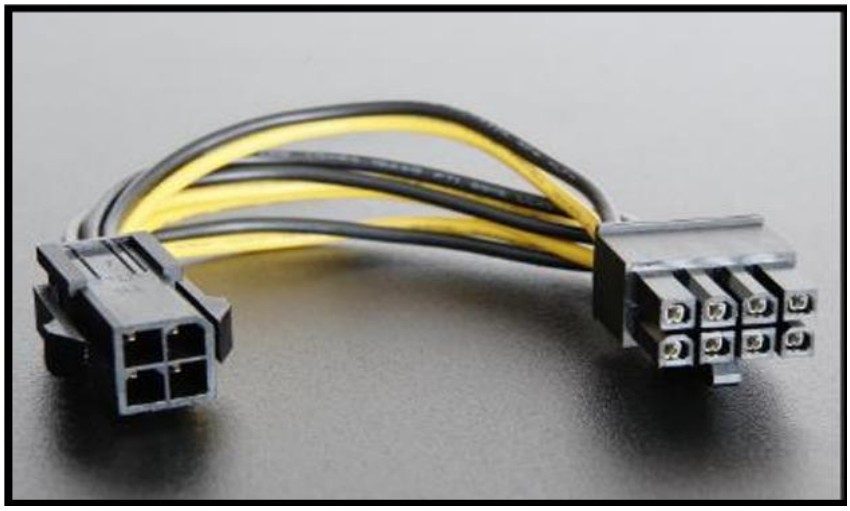
4. **Power Connectors**

Function: Supply power to the motherboard
Specifications:

• 24-pin main power connector
• 4-pin / 8-pin CPU power connector



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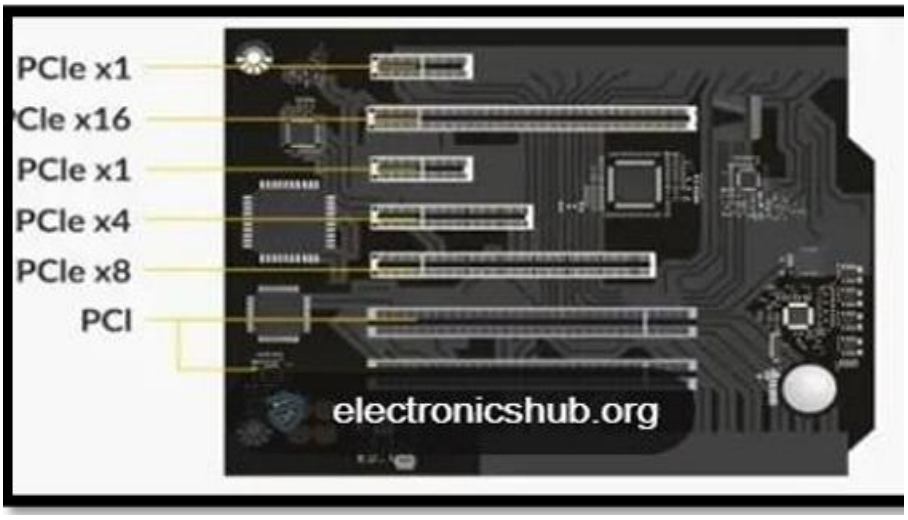
5. Expansion Slots (PCI / PCIe)

Function: Used to install additional cards
Specifications:
• PCIe x16 → Graphics card
• PCIe x1 → Sound card / Network card
• Versions: PCIe 3.0, 4.0, 5.0



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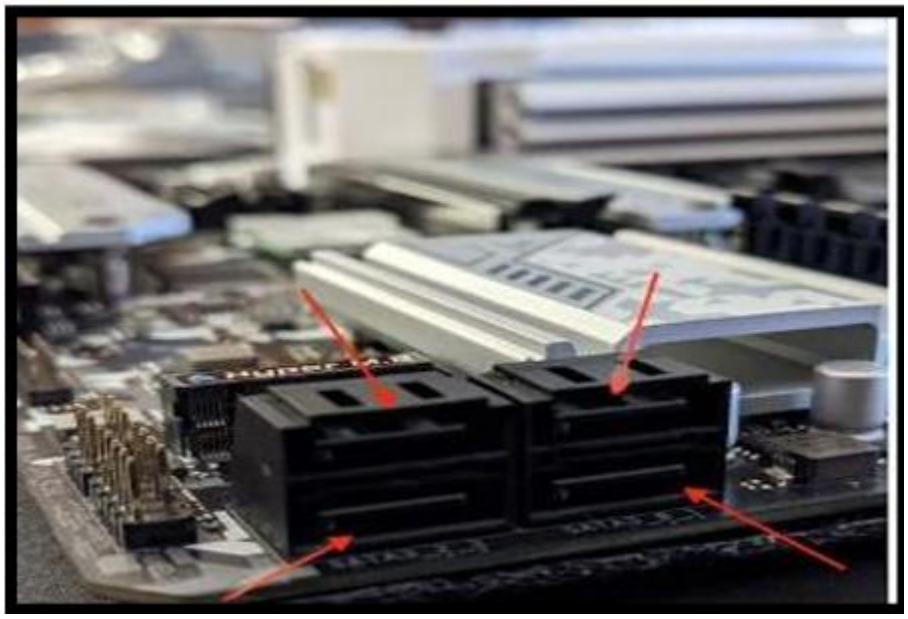
6. Storage Connectors

Function: Used to connect HDD and SSD
Specifications:

• SATA ports → HDD / SSD
• M.2 slot → NVMe SSD (high speed)



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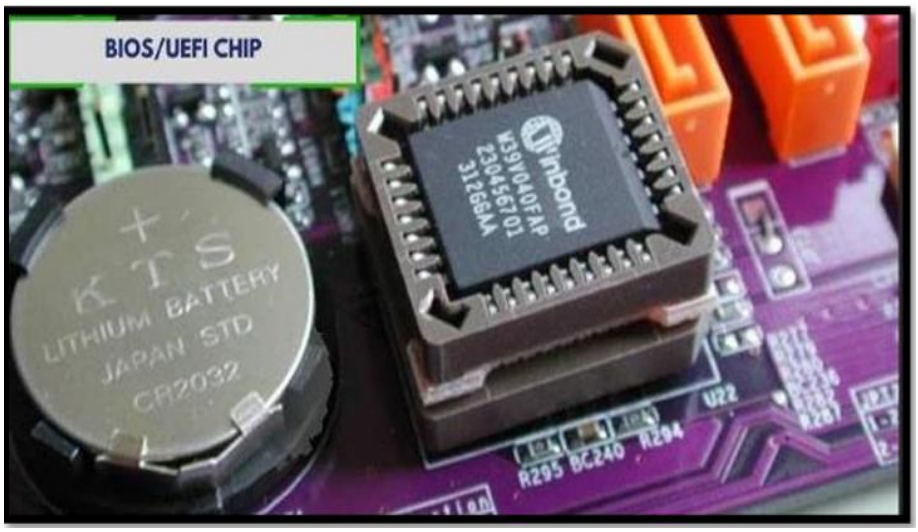
7. BIOS / UEFI Chip

Function: Starts the system (booting)
Specifications:
• Flash memory chip
• Stores hardware configuration settings



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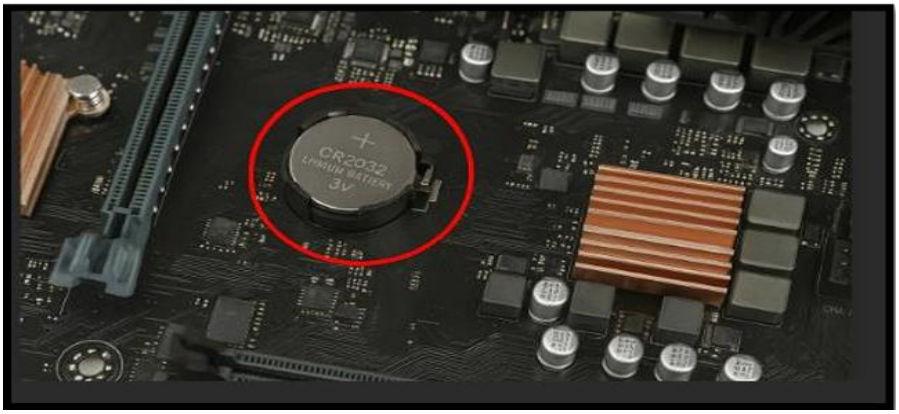
8. CMOS Battery

Function: Saves date, time, and BIOS settings
Specifications:
• Type: CR2032
• Voltage: 3 volts



9. **Back Panel Ports**

Function: Used to connect external devices

Specifications:

• USB 2.0 / USB 3.0 / Type-C

• HDMI / VGA / DisplayPort

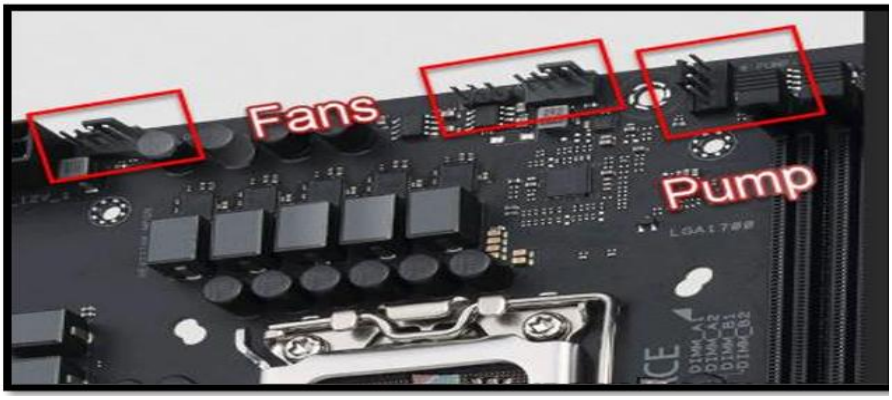
• LAN and Audio ports

10. **Fan and Front Panel Headers**

Function: Used for cooling and cabinet controls

Specifications:

• CPU fan header

• System fan header

• Power button and reset button pins